

Week 2: Prompt Engineering & LangChain

Learning Topics

System, User & Assistant Prompts

Prompt Templates

Role Prompting

Chain-of-Thought Prompting

Structured Output (JSON)

Function Calling (Introduction)

LangChain Architecture

Models

Prompt Templates in LangChain

Output Parsers

Chains

Weekly Task

Build an **AI Customer Support Assistant** using **LangChain** that can answer frequently asked customer questions.

The application should use any one of the following LLM APIs: **OpenAI, Gemini, Claude, or any free LLM** supported by LangChain.

The assistant should provide accurate, professional, and context-aware responses through either a **Command-Line Interface (CLI)** or a **Web Interface**.

Expected Deliverables

Complete source code

GitHub repository

README.md with setup instructions

requirements.txt

Environment variables configured using a `.env` file

Working AI Customer Support Assistant built using LangChain

Implementation of Prompt Templates

Sample customer questions and AI-generated responses

Submission

Submit the GitHub repository link before the assigned deadline. Ensure the repository includes all required files, proper documentation, and clear setup instructions.